LIBRARY MANAGEMENT SYSTEM USING PHP & MYSQL

Project Submitted by

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INTRODUCTION

Abstract

- The Library Management System using PHP and MySQL is a web-based application developed to automate and manage library operations efficiently. This system is designed to replace the traditional manual library system with a computerized solution that ensures accuracy, speed, and ease of access to library resources. PHP is used as the server-side scripting language, while MySQL serves as the backend database to store and manage library data securely.

- This system reduces paperwork, minimizes human errors, and improves data consistency by maintaining all records in a centralized database. Features such as automatic fine calculation, user authentication, report generation, and role-based access control enhance the reliability and security of the application.

Objectives

- The main objective of this project is to provide an efficient platform for managing book records, user information, and library transactions such as book issue, return, and renewal. The system allows administrators and librarians to add, update, delete, and search books, manage student or member records, and monitor issued books in real time. Users can log in to search for available books, view issued books, and check due dates through a user-friendly web interface.

- This system reduces paperwork, minimizes human errors, and improves data consistency by maintaining all records in a centralized database. Features such as automatic fine calculation, user authentication, report generation, and role-based access control enhance the reliability and security of the application.

PROJECT MODULES

- In this project, we use PHP and MySQL database. This application is basically managed by the administrator and administrator has the following rights.
  - Students
  - Admin

1. Student Module

- Student can register yourself and after registration they will get studentid
- After login student can view own dashboard.
- Stduents can also view the listed books in the library.
- Student can update own profile.
- Student can view issued book and book return date-time.
- Student can also change own password.
- Student can also recover own password.

ADMIN MODULE

- Admin Dashboard
- Admin can add/update/ delete category
- Admin can add/update/ delete author
- Admin can add/update/ delete books
- Admin can issue a new book to student and also update the details when student return book
- Admin can search student by using their student ID
- Admin can also view student details
- Admin can change own password

Proposed System

- The proposed Library Management System using PHP and MySQL is a fully web-based, automated solution designed to manage all library operations efficiently. The system replaces the existing manual or semi-computerized processes with a centralized digital platform accessible through a web browser.
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- In this system, all library data such as book details, member information, issue/return records, and fine details are stored in a MySQL database. PHP is used to handle server-side logic, user authentication, and interaction with the database. The system supports multiple user roles including Administrator, Librarian, and User (Student/Member), each with defined access permissions

Existing System

- The existing library system is mostly based on manual or semi-computerized processes where records are maintained using registers, files, or basic spreadsheet applications. In this system, all library activities such as book entry, member registration, book issuing, and returning are handled manually by library staff.

- Book details, student information, and transaction records are recorded in physical registers, which makes data retrieval slow and difficult. Searching for a particular book or checking its availability requires manual verification, consuming a significant amount of time. Due date tracking and fine calculation are also done manually, which often leads to calculation errors and inconsistencies.

- The existing system does not provide real-time access to information for users. Students or members must physically visit the library to know the availability of books or the status of issued books. Data security is weak, as records are prone to loss, damage, or unauthorized access.

Hardware & Software Requirements

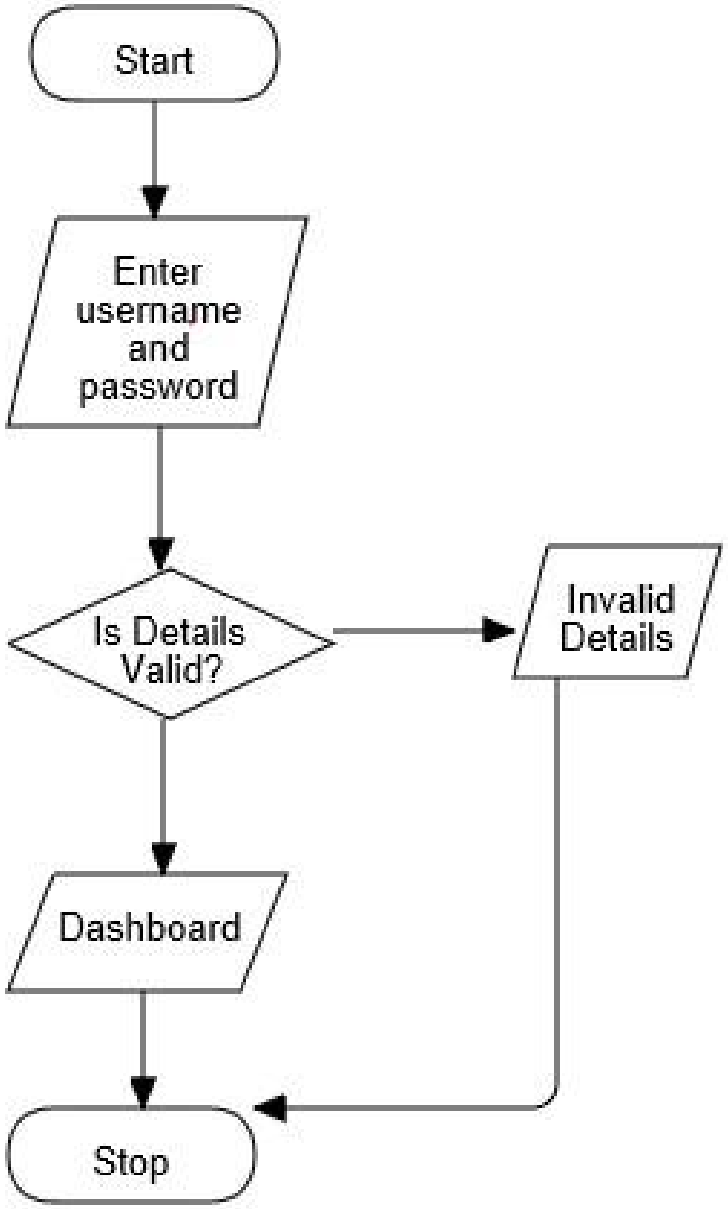
Software Requirements

Platform	:	HTML5, PHP3
Database	:	My-SQL 5.6.8
Software	:	WAMPSERVER64, Dream-viewer 8
Media Package	:	J-Query, JavaScript
Protocol	:	FTP, HTTPS

Hardware Requirements

- Processor - Pentium IV
- Hard disk - 80GB Hard disk
- RAM - 1GB RAM
- OS - Windows XP/7/8.1/10/11
- Internet Usage - 1.2 Mbps

FLOW CHART & CLASS DIAGRAM OF LMS



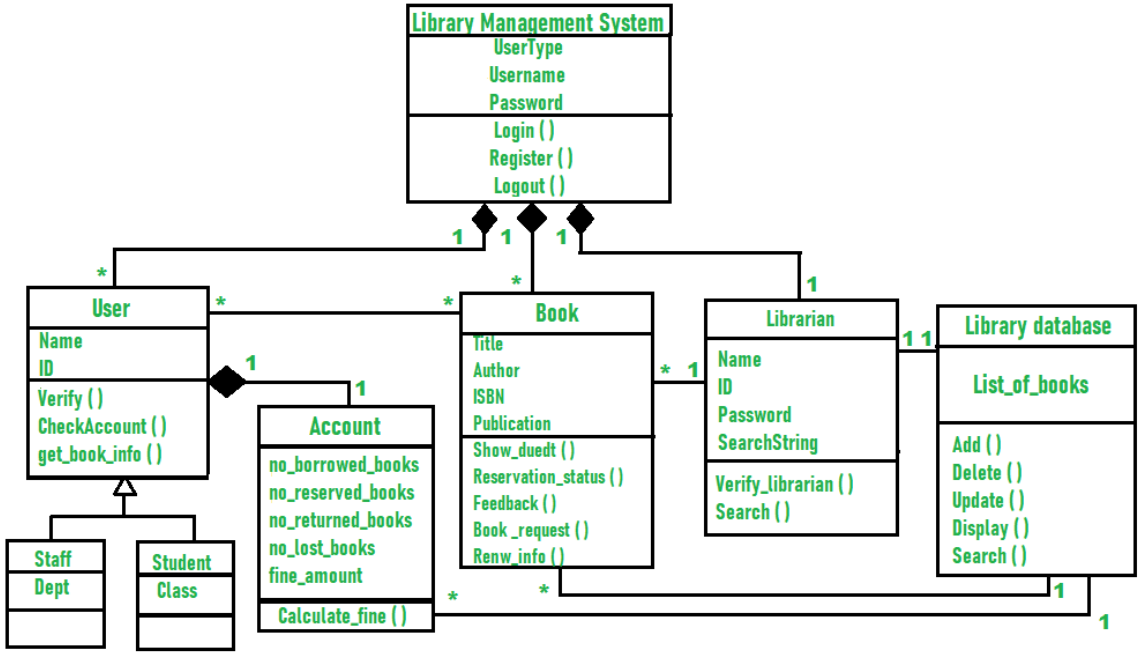
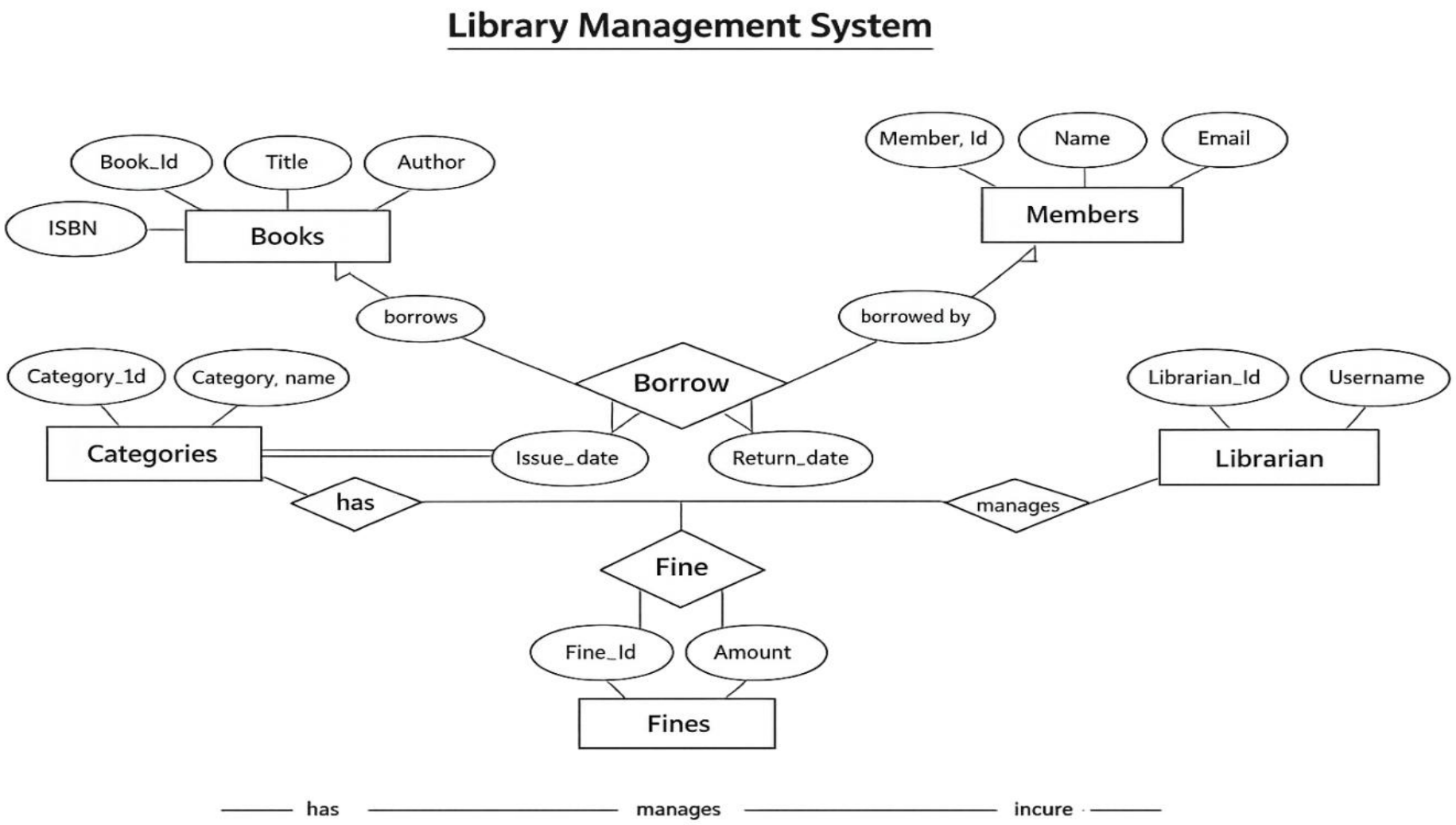


TABLE STRUCTURE OF LMS

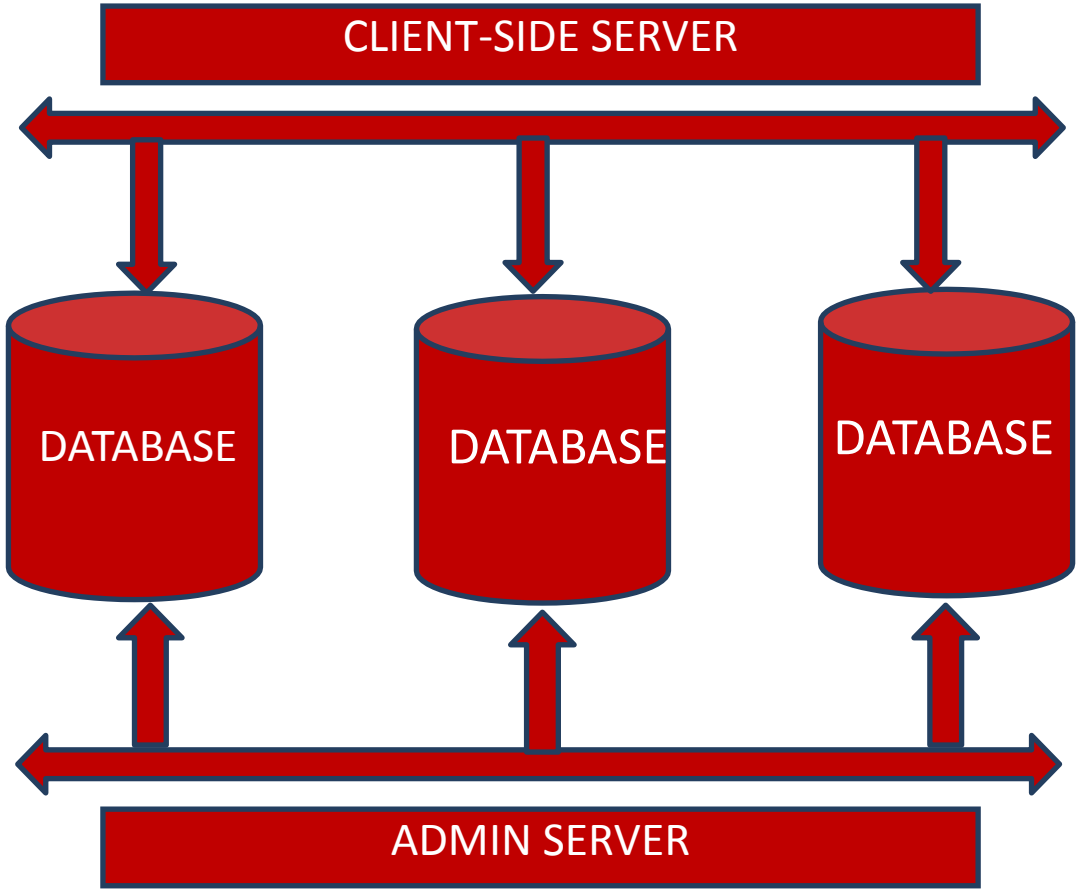


Database Model Diagram

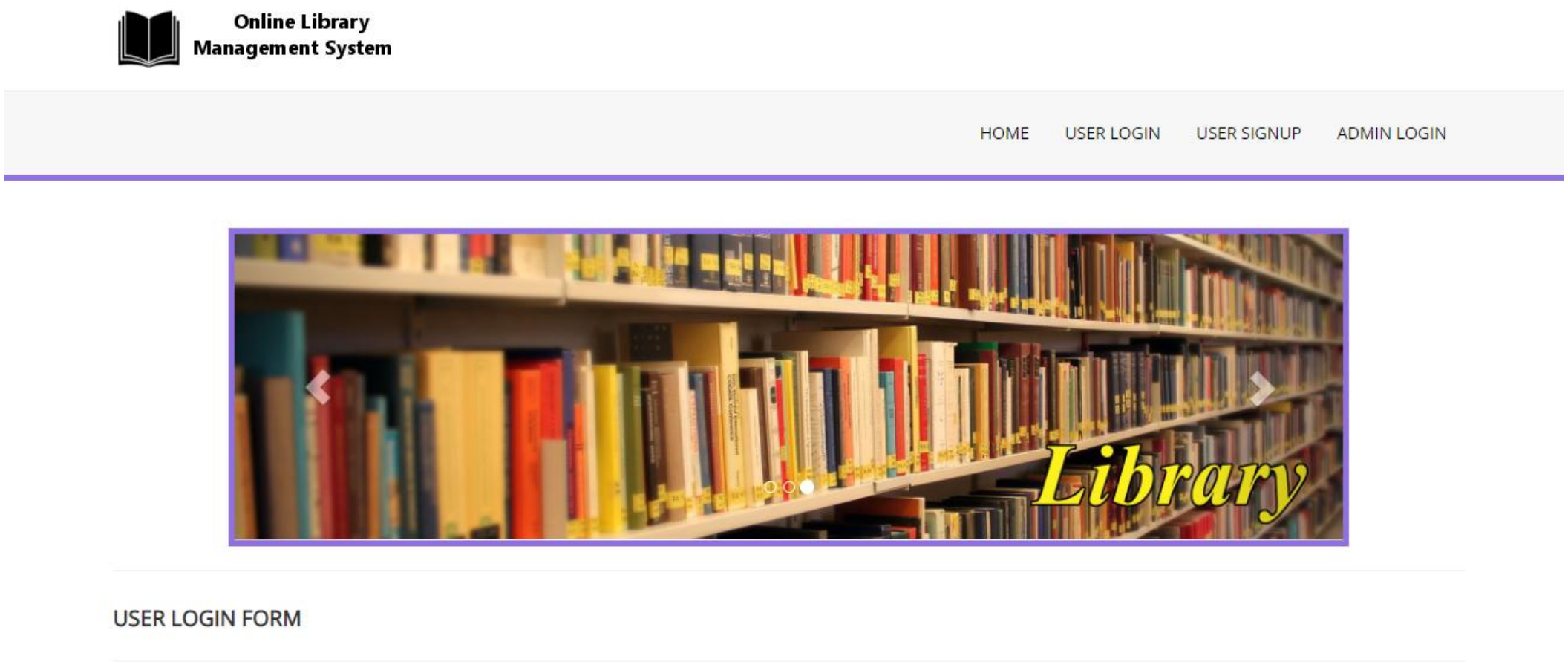
LOCALSERVER

SOFTWARE BASED DATA CONNECTION SYSTEMS ARE USED.

SO, NO NEED INTERNET CONNECTION IN THIS PROJECT



Project Screenshot



USER LOGIN FORM

LOGIN FORM

Enter Email id

Password

Forgot Password

LOGIN | Not Register Yet

Online Library Management System

HOME USER LOGIN USER SIGNUP ADMIN LOGIN

USER SIGNUP

SINGUP FORM

Enter Full Name

Mobile Number :

Enter Email

Enter Password

Confirm Password

Register Now

Online Library
Management System

HOME USER LOGIN USER SIGNUP ADMIN LOGIN

ADMIN LOGIN FORM

LOGIN FORM

Enter Username

Password

LOGIN

Online Library Management System

LOG ME OUT

DASHBOARD | CATEGORIES | AUTHORS | BOOKS | ISSUE BOOKS | REG STUDENTS | CHANGE PASSWORD

ADMIN DASHBOARD

9
Books Listed

2
Books Not Returned Yet

7
Registered Users

11
Authors Listed

6
Listed Categories

Online Library Management System

LOG ME OUT

DASHBOARD | CATEGORIES | AUTHORS | BOOKS | ISSUE BOOKS | REG STUDENTS | CHANGE PASSWORD

MANAGE BOOKS

Books Listing

10 records per page

Search:

#	Book Name	Category	Author	ISBN	Price	Action
1	PHP And MySql programming	Technology	Anuj kumar	222333	20.00	Edit Delete
2		Technology	Anuj kumar	9350237695	455.00	Edit Delete

Online Library Management System

LOG ME OUT

DASHBOARD | CATEGORIES | AUTHORS | BOOKS | ISSUE BOOKS | REG STUDENTS | CHANGE PASSWORD

MANAGE ISSUED BOOKS

Issued Books

10 records per page

Search:

#	Student Name	Book Name	ISBN	Issued Date	Return Date	Action
1	John Doe	C++: The Complete Reference, 4th Edition	007053246X	2024-02-01 11:15:57	2024-02-04 12:03:08	Edit
2	John Doe	PHP And MySql programming	222333	2024-02-01 11:15:57	Not Return Yet	Edit
3	test	ASP.NET Core 5 for Beginners	GBSJ36344563	2024-02-01 11:15:57	2024-02-04 12:03:08	Edit
4	test	C++: The Complete Reference, 4th Edition	007053246X	2024-02-01 11:15:57	2024-02-04 12:03:08	Edit
5	Anuj kumar	PHP And MySql programming	222333	2024-02-01 11:15:57	2024-02-04 12:03:08	Edit

Online Library Management System

LOG ME OUT

DASHBOARD | CATEGORIES | AUTHORS | BOOKS | ISSUE BOOKS | REG STUDENTS | CHANGE PASSWORD

MANAGE AUTHORS

Authors Listing

10 records per page

Search:

#	Author	Creation Date	Updation Date	Action
1	Anuj kumar	2024-01-25 12:53:03	2024-02-04 12:04:19	Edit Delete
2	Chetan Bhagatt	2024-01-25 12:53:03	2024-02-04 12:04:26	Edit Delete
3	Anita Desai	2024-01-25 12:53:03	2024-02-04 12:04:26	Edit Delete
4	HC Verma	2024-01-25 12:53:03	2024-02-04 12:04:26	Edit Delete
5	R.D. Sharma	2024-01-25 12:53:03	2024-02-04 12:04:26	Edit Delete

Online Library Management System

LOG ME OUT

DASHBOARD | CATEGORIES | AUTHORS | BOOKS | ISSUE BOOKS | REG STUDENTS | CHANGE PASSWORD

ADD CATEGORY

Category Info

Category Name

Science

Status

- (•) Active
- ( ) Inactive

Create

Online Library Management System

LOG ME OUT

DASHBOARD | CATEGORIES | AUTHORS | BOOKS | ISSUE BOOKS | REG STUDENTS | CHANGE PASSWORD

ADD AUTHOR

Author Info

Author Name

Add

ADD BOOK

Book Info

Book Name*

Category*

Select Category

Author*

Select Author

ISBN Number*

An ISBN is an International Standard Book Number.ISBN Must be unique

Price*

Book Picture*

Choose file No file chosen

Book Quantity*

Submit

CONCLUSION

- The Library Management System project successfully automates the major activities of a traditional library such as book management, member registration, issue/return tracking, and fine calculation. By using PHP for server-side processing and MySQL for database management, the system provides a reliable, secure, and user-friendly platform for handling library operations.

- Overall, the project demonstrates how web technologies can be effectively used to build real-time management systems. The application is scalable and can be enhanced further by adding modules such as online reservations, email notifications, barcode scanning, and role-based access control.

- In conclusion, the Library Management System offers an efficient, cost-effective, and modern solution for managing library resources.

Thank You!

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